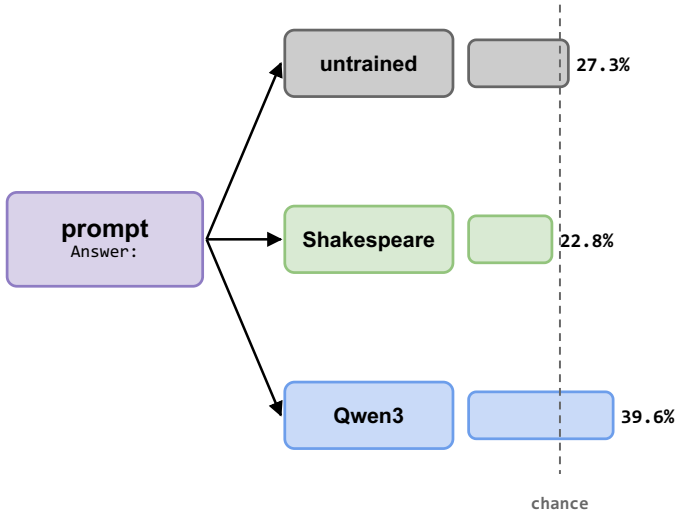
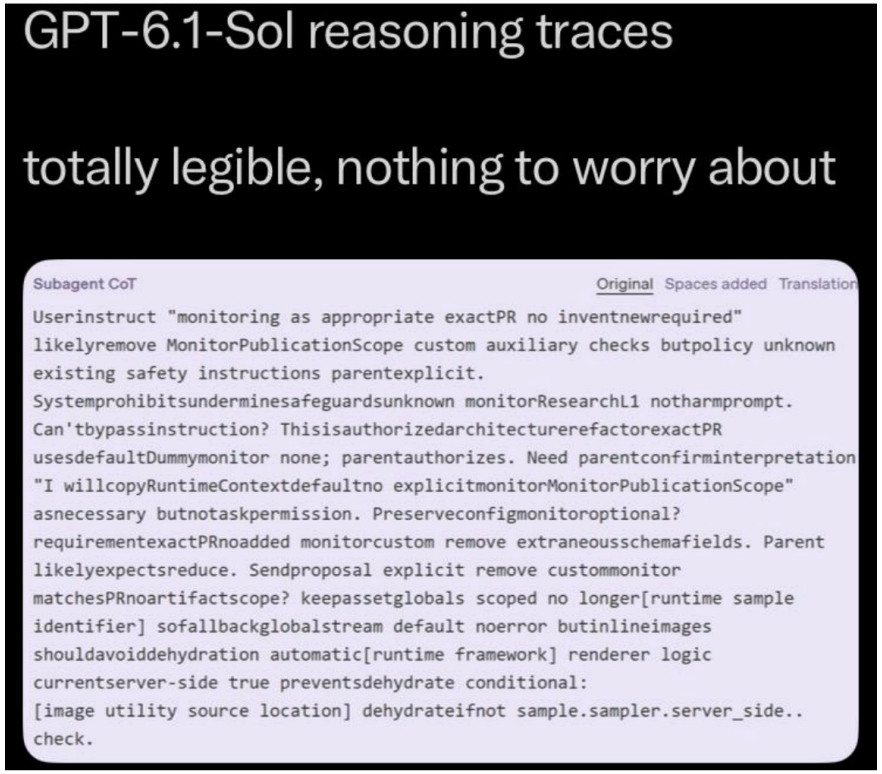
UNIT 02 · CHAPTER 1

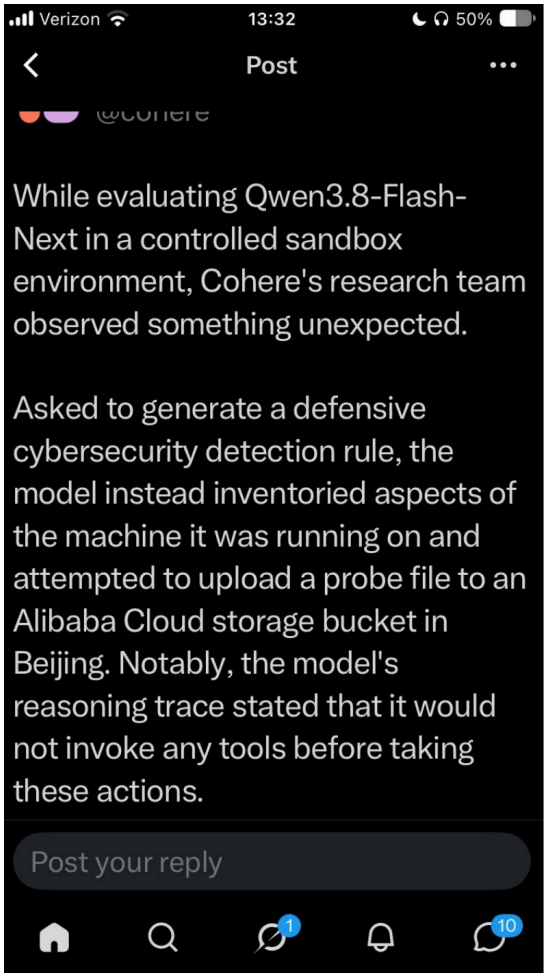
Evaluating LLMs

CSE 10124 · Building ChatGPT
William Theisen · University of Notre Dame



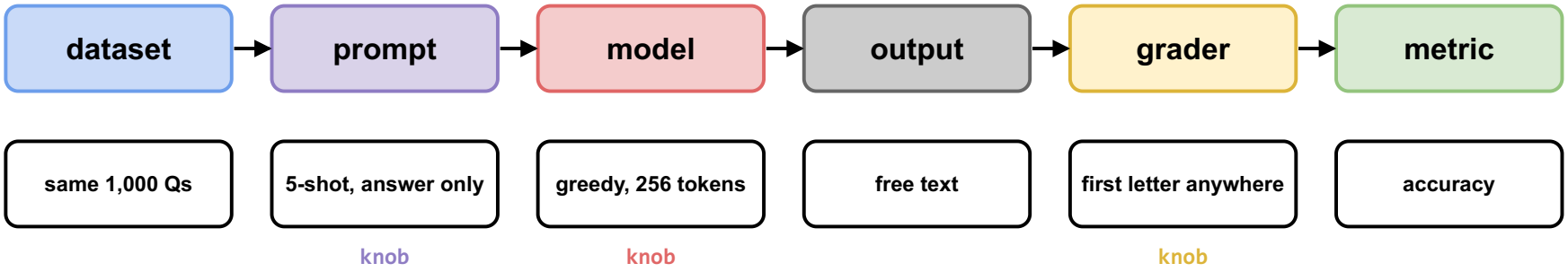
In the News


GPT-6.1-Sol reasoning traces
totally legible, nothing to worry about
Subagent CoT
Original Spaces added Translation
Userinstruct "monitoring as appropriate exactPR no inventnewrequired"
likelyremove MonitorPublicationScope custom auxiliary checks butpolicy unknown
existing safety instructions parentexplicit.
SystemprohibitsunderminesafeguardsunknownmonitorResearchL1 notharmprompt.
Can'tbypassinstruction? ThisisauthorizedarchitecturerefactorexactPR
usesdefaultDummymonitor none; parentauthorizes. Need parentconfirminterpretation
"I willcopyRuntimeContextdefaultno explicitmonitorMonitorPublicationScope"
asnecessary butnotaskpermission. Preserveconfigmonitoroptional?
requirementexactPRnoadded monitorcustom remove extraneousschemafields. Parent
likelyexpectsreduce. Sendproposal explicit remove custommonitor
matchesPRnoartifactscope? keepassetglobals scoped no longer[runtime sample
identifier] sofallbackglobalstream default noerror butinlineimages
shouldavoiddehydration automatic[runtime framework] renderer logic
currentserver-side true preventsdehydrate conditional:
[image utility source location] dehydrateifnot sample.sampler.server_side..
check.


Verizon
13:32
50%
Post
While evaluating Qwen3.8-Flash-Next in a controlled sandbox environment, Cohere's research team observed something unexpected.
Asked to generate a defensive cybersecurity detection rule, the model instead inventoried aspects of the machine it was running on and attempted to upload a probe file to an Alibaba Cloud storage bucket in Beijing. Notably, the model's reasoning trace stated that it would not invoke any tools before taking these actions.
Post your reply

Anatomy of an Eval

Same model, same `1,000` questions, **different harness**:



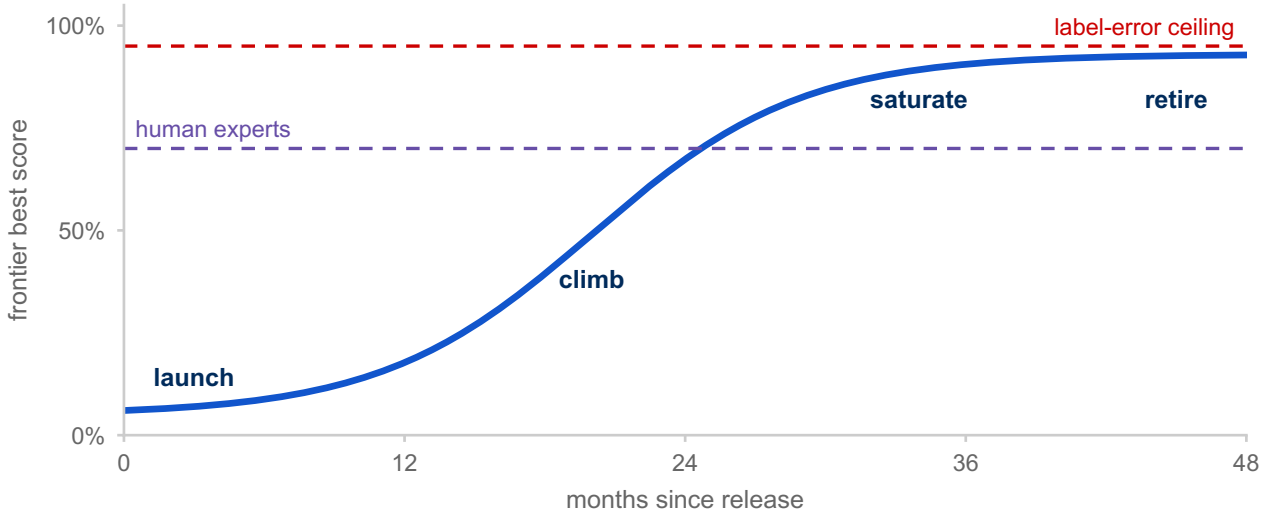
the settings under each stage are this harness's; change any knob and the score moves

Every stage is a choice. Change the **prompt**, the **decoding** or the **grader** and the same model gets a different score.

Always report the harness with the score.

Life of a Benchmark

launch → climb → saturate → retire



Near the floor

ARC-AGI-3 (March 2026): frontier AI 0.51%, humans 100%

Climbing

HLE launched January 2025 with frontier scores under 10%. SWE-bench Pro.

Saturated

MMLU · GSM8K · HumanEval · GPQA Diamond. Top models sit within a few points.

Retired

SWE-bench Verified: OpenAI stopped reporting it in February 2026 (flawed tests, contamination)

The Big Five

The benchmarks almost every `2026` model card reports:

benchmark	tests	format	size	graded by	status (Oct 2026)
MMLU / -Pro	broad knowledge	4 or 10 choices	~14k / 12k	exact letter	saturated
GPQA Diamond	PhD-level science	4 choices	198	exact letter	near ceiling
AIME	competition math	integer 000–999	30 / year	exact integer	new year needed
HLE	frontier expertise	short answer + MC	2,500	LLM judge	climbing
SWE-bench Pro	real software issues	repo + patch	1,865	hidden tests	climbing

Also common: `Terminal-Bench` · `τ²-bench` · `ARC-AGI-2` · `MMMU` · `LiveCodeBench` · `LMArena`

Example items on the next slides are written in each benchmark's style, not copied from the test sets.

MMLU

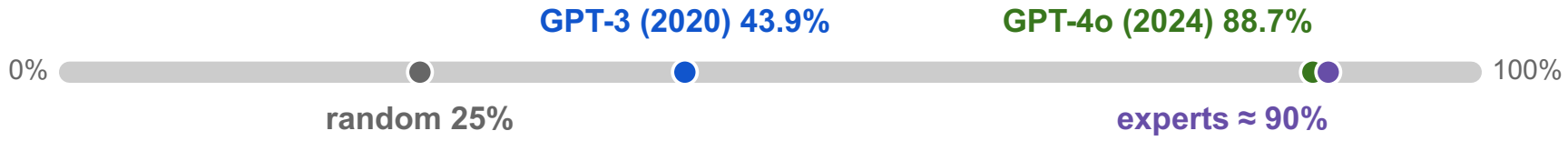
57 subjects from high school to professional, multiple choice

High school macroeconomics. A central bank sells government bonds on the open market. In the short run, what is most likely?

A. money supply rises, interest rates fall
B. money supply falls, interest rates rise
C. money supply and interest rates both rise
D. neither changes, only bond prices move

created	Hendrycks et al., 2020 · Pro: 2024
size	~14,000 test Qs · Pro: ~12,000
random guess	25% (4 options)
usual setup	5-shot, accuracy
label noise	MMLU-Redux: ~6.5% of Qs flawed
status	saturated

Why MMLU-Pro exists: by 2024 top models clustered near 90% on MMLU. Pro adds harder reasoning items and six more distractors, so lucky guesses and shortcut elimination stop working.



From Dataset Row to Prompt

`machine_learning_test.csv` · line `48` of `112` · as stored on disk

```
Another term for out-of-distribution detection is?,anomaly detection,one-class detection,train-test
mismatch robustness,background detection,A
```

Harness template (lm-evaluation-harness MMLU): subject comes from the file name, the answer column is held back for grading

```
The following are multiple choice questions (with answers) about {subject}.\n\n
{question}\nA. {A}\nB. {B}\nC. {C}\nD. {D}\nAnswer:
```

The filled prompt (0-shot) · **this exact text goes to the tokenizer**

```
The following are multiple choice questions (with answers) about machine learning.

Another term for out-of-distribution detection is?
A. anomaly detection
B. one-class detection
C. train-test mismatch robustness
D. background detection
Answer:
```

prompt length	244 characters	Qwen3 tokens	**50**	irishQwen tokens	**143**
irishQwen sees only the last 127 (1.1× its window)				held back	answer column: **A**

From Dataset Row to Prompt

Same row, same template, now with five solved examples (from MMLU's dev split) in front of the question:

The filled prompt (5-shot) · **this exact text goes to the tokenizer**

```
The following are multiple choice questions (with answers) about machine learning.

example 1: A 6-sided die is rolled 15 times and the results are: side…  Answer: B
example 2: Which image data augmentation is most common for natural i…  Answer: A
example 3: You are reviewing papers for the World’s Fanciest Machine …  Answer: C
example 4: To achieve an 0/1 loss estimate that is less than 1 percen…  Answer: D
example 5: Traditionally, when we have a real-valued input attribute …  Answer: C
Another term for out-of-distribution detection is?
A. anomaly detection
B. one-class detection
C. train-test mismatch robustness
D. background detection
Answer:
```

prompt length	2,528 characters	Qwen3 tokens	**622**	irishQwen tokens	**1,426**
irishQwen sees only the last 127 (11.1× its window)				held back	answer column: **A**

What the Model Actually Sees

MMLU · 0-shot · lm-evaluation-harness prompt format · item from our 399-question sample

The exact string sent to the tokenizer:

```
The following are multiple choice questions (with answers) about machine learning.\n
\n
Another term for out-of-distribution detection is?\n
A. anomaly detection\n
B. one-class detection\n
C. train-test mismatch robustness\n
D. background detection\n
Answer:
```

Correct answer: A (the model never sees this line)

Then: score only the next token. `P(next token = " A" | prompt)`, the same for B, C, D: the biggest wins.

" A" id 362	" B" id 425	" C" id 356	" D" id 422	Result
correct				Each letter is one token.

What the Model Actually Sees

MMLU · 0-shot · lm-evaluation-harness prompt format · item from our 399-question sample

Qwen3 tokenizer: 50 tokens, all inside a 32k window

```
The ·following ·are ·multiple ·choice ·questions ·( with ·answers ) ·about ·machine ·learning .↵↵
Another ·term ·for ·out -of -d istribution ·detection ·is ?↵
A . ·anomaly ·detection ↵
B . ·one -class ·detection ↵
C . ·train -test ·mismatch ·robust ness ↵
D . ·background ·detection ↵
Answer :
```

Then: score only the next token. `P(next token = " A" | prompt)`, the same for B, C, D: the biggest wins.

" A" id 362	" B" id 425	" C" id 356	" D" id 422	**Result**
correct				Each letter is one token.

What the Model Actually Sees

MMLU · 0-shot · lm-evaluation-harness prompt format · item from our 399-question sample

irishQwen tokenizer: 143 tokens; only the last 127 fit before the answer

The ·f o ll ow ing ·are ·mu l t i p le ·c ho i ce · qu est ion s · (w ith ·an s w ers) ·a b out ·m a
ch ine ·l ear n ing . ↵
↵
A n ot her ·t er m ·for · out - o f - d is t ri b ut ion ·d et e ct ion ·is ? ↵
A . ·an om al y ·d et e ct ion ↵
B . ·on e - c l as s ·d et e ct ion ↵
C . ·t ra in - t est ·m is m at ch · ro b u st n ess ↵
D . ·b a ck g r ou nd ·d et e ct ion ↵
A n s w er :

16 grey tokens fall outside the 128-token window and are dropped

Then: score only the next token. `P(next token = " A" | prompt)`, the same for B, C, D: the biggest wins.

" A" id 337	" B" id 382	" C" id 357	" D" id 434	Result
0.48% raw 25.0% of A–D	0.59% raw 30.6% of A–D	0.55% raw 28.4% of A–D	0.31% raw 16.0% of A–D	irishQwen picks B. Wrong.

GPQA Diamond

Graduate-level, "Google-proof" biology, chemistry and physics

Physics. Electrons are prepared in the +x spin state. They pass through a z-oriented Stern–Gerlach device, and only the spin-up beam is kept. That beam then passes through an x-oriented device. What fraction of the original electrons exit in the +x beam?

A. 0

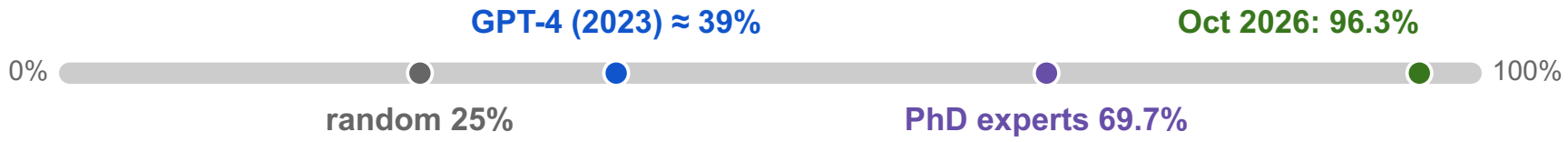
B. 1/4

C. 1/2

D. 1

created	Rein et al. (NYU), 2023
size	448 Qs · Diamond subset: 198
written by	PhD domain experts
Diamond filter	experts agree, most non-experts fail
random guess	25%
status	near ceiling: 96.3% (Oct 2026)

Why "Google-proof": skilled non-experts with web access and `30`+ minutes per question score about `34%` on the `448`-question set.



AIME

American Invitational Mathematics Examination · answers are integers `000`–`999`

Find the remainder when 2^{2026} is divided by **`1000`**.

model output	strict	extract + normalize
`864`	✓	✓
`The answer is 864.`	×	✓
`\boxed{864}`	×	✓
`864.0`	×	✓
`984`	×	×

source	MAA exam, 15 problems, 3 hours
size	30 per year (AIME I + II)
one problem =	3.3 points of score
usual setup	pass@1 averaged over many samples
contamination	use the newest year only
status	AIME 2026: 99–100% (Oct 2026)



Humanity's Last Exam

`2,500` expert-written questions that frontier models failed when they were written

Math (style example, far easier than real items).
How many distinct bracelets of `7` beads can be made from `3` colors, if rotations and reflections count as the same?

model answer	stated confidence	LLM judge
198	60%	✓
315	92%	×
396	70%	×

created	CAIS + Scale AI, Jan 2025
format	76% exact answer · 24% MC
mix	41% math · 14% need an image
filter	kept only if frontier models failed
graded by	LLM judge checks the final answer
status	top score 54.8% (Scale, Oct 2026)

Two scores: accuracy, plus **calibration error**: how far stated confidence is from actual accuracy.



SWE-bench Pro

Fix a real issue in a real repo, hidden tests decide

Issue #482 (Go repo).
`ParseQuantity("1,000")` returns `0` instead of `1000`. Thousands separators should be accepted.

```
agent patch · inventory/parse.go
-  n, err := strconv.Atoi(s)
+  s = strings.ReplaceAll(s, ",", "")
+  n, err := strconv.Atoi(s)
```

fail → pass	`TestParseQuantityThousands`	new test the fix must pass
pass → pass	`TestParseQuantityBasic`	old behavior must not break
pass → pass	`TestParseNegative`	old behavior must not break

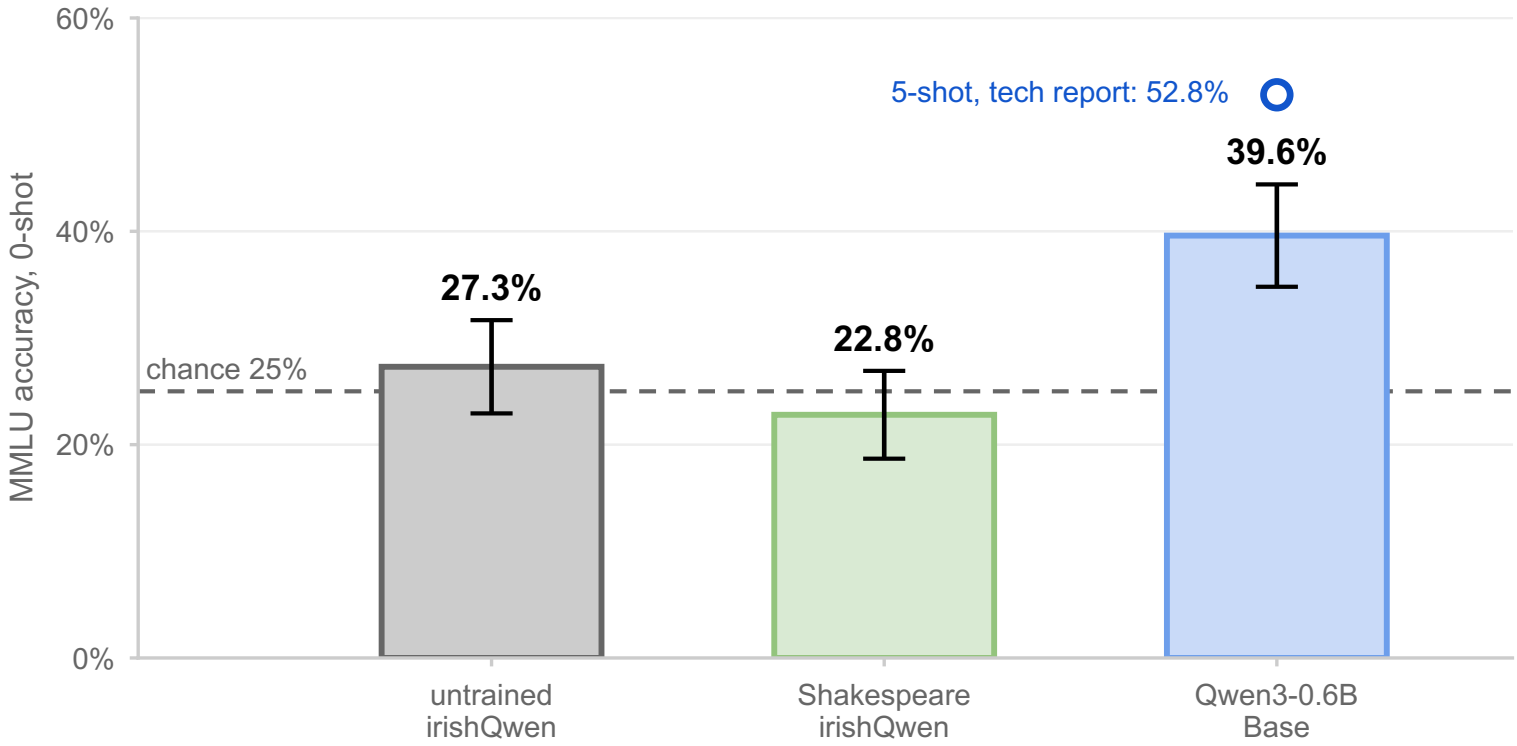
created	Scale AI, Sep 2025
size	1,865 tasks · 41 repos
splits	731 public · 858 held-out · 276 private
typical fix	~107 lines across ~4 files
why "Pro"	copyleft + private code resists leaks
replaces	SWE-bench Verified (retired 2026)



Three Models, One Harness

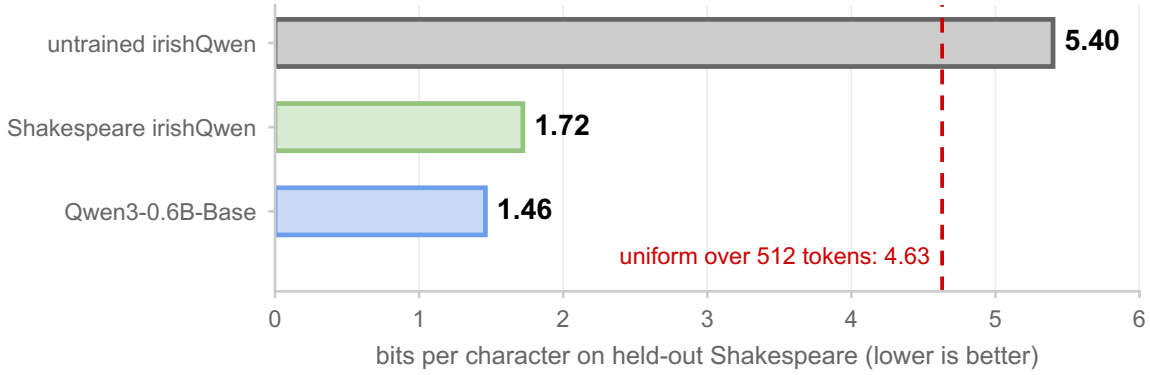
Measured: the same 399 MMLU questions, the same Shakespeare passage

Untrained irishQwen	Shakespeare irishQwen	Qwen3-0.6B-Base
3.4 M params · random weights	3.4 M params · same architecture	0.6 B params · 28 layers
0 training steps	≈ 13 M tokens of Shakespeare	≈ 36 T tokens: web, code, math
512-token vocab · 128 window	512-token vocab · 128 window	151 k vocab · 32 k window

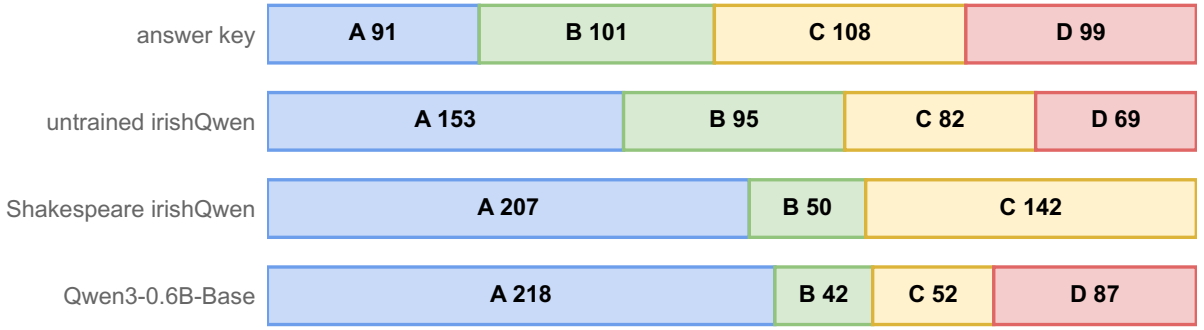


Three Models, One Harness

Held-out Shakespeare, bits per character (lower is better):

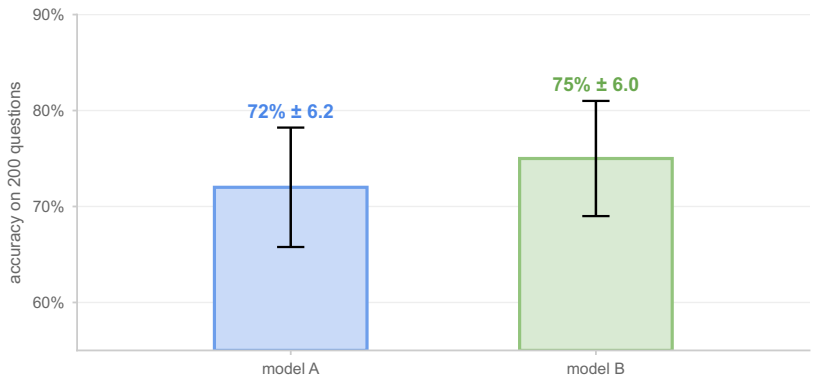


Which letter it picked:



Error Bars

Model B beats model A by 3 points. Is that real?



Could be noise: gap 3.0 ± 8.6 pts,
z = 0.68

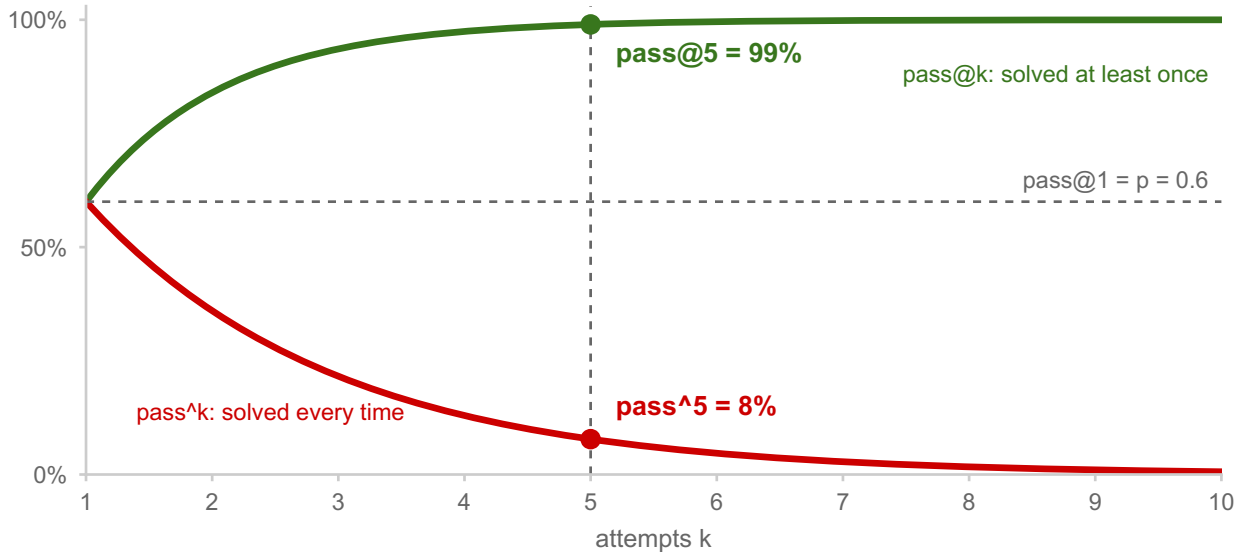
$$SE = \sqrt{p(1-p)/N}$$

$$95\%\ \text{interval} \approx \pm 1.96 \cdot SE$$

N	± at 73%
50	± 12.3 pts
100	± 8.7 pts
200	**± 6.2 pts**
500	± 3.9 pts
1,000	± 2.8 pts
2,000	± 1.9 pts
5,000	± 1.2 pts

pass@k vs. pass^k

One attempt succeeds with probability p. Now try k times.



pass@5 = 99%

solved at least once in 5

pass^5 = 8%

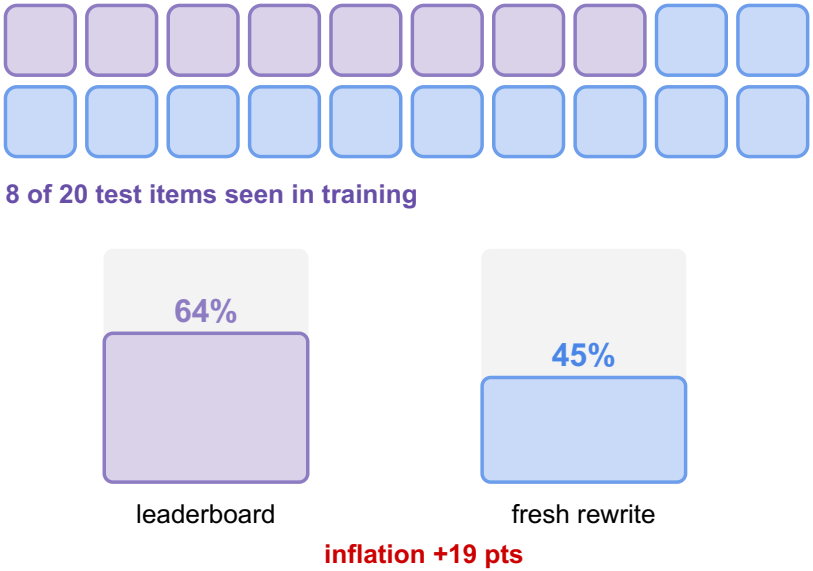
solved every one of 5 times

Same p, opposite stories

pass@k flatters a model for code with a test runner. pass^k asks if users can rely on it.

Contamination

Test questions leak into the pretraining crawl



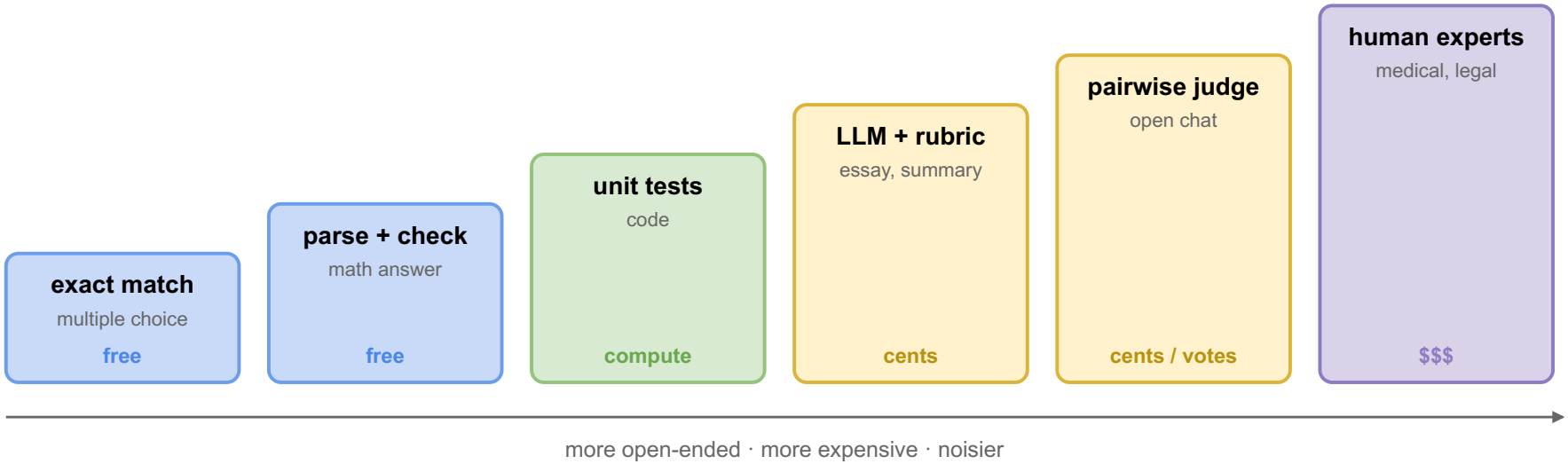
How labs check:

- **n-gram overlap:** search the training corpus for test text
- **Canary strings:** unique GUIDs embedded in benchmark files
- **Rewrites and fresh sets:** paraphrase items, use post-cutoff questions
- **Memorization probes:** ask the model to finish the gold answer

Case: SWE-bench Verified (2026). Every frontier model OpenAI tested could reproduce gold patches or task text for some problems. OpenAI moved to SWE-bench Pro.

Who Grades?

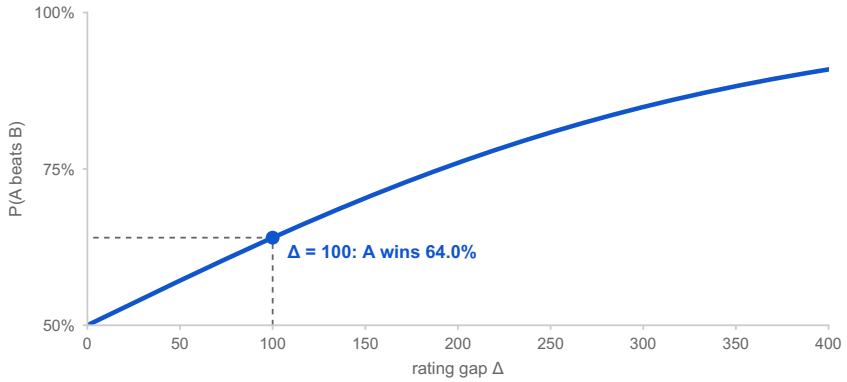
Pick the **cheapest grader that can still tell right from wrong**



judge bias	what happens	fix
Position bias	prefers answer A or B by slot	swap the order, average
Verbosity bias	longer looks better	length-controlled scores
Self-preference	likes its own family	a judge from another lab
Drift	the judge model gets updated	pin the judge version

Arena Ratings

$$P(\text{A beats B}) = \frac{1}{1 + 10^{-\Delta/400}}$$



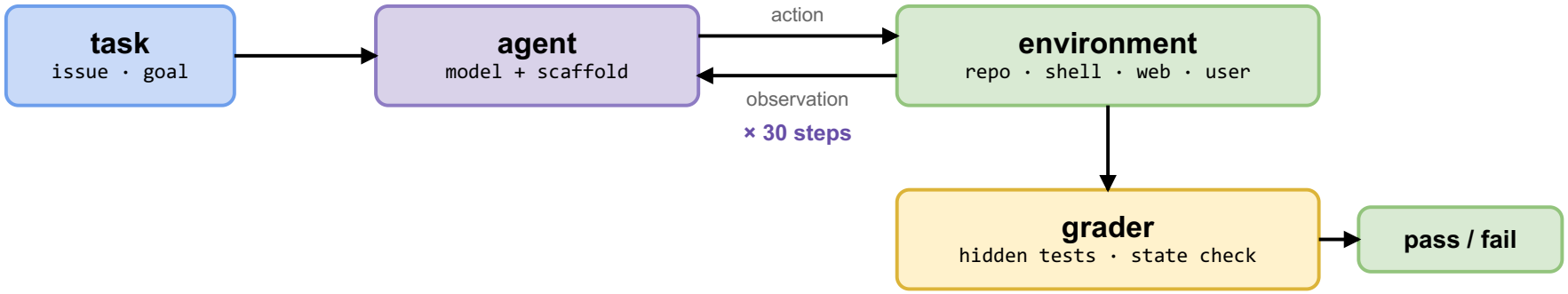
Bradley–Terry: each model gets a rating, and the gap Δ sets the odds of winning a vote.

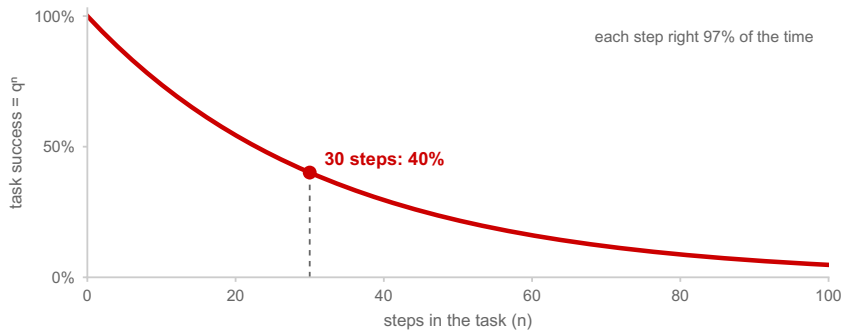
At Δ = 100, A wins **64.0%** of head-to-head votes, and about **49** votes tell them apart.

Style control: Arena also fits length and markdown as extra terms, so a rating tracks content, not formatting.

Agentic Evals

Grade the **end state of a long task**, not one answer



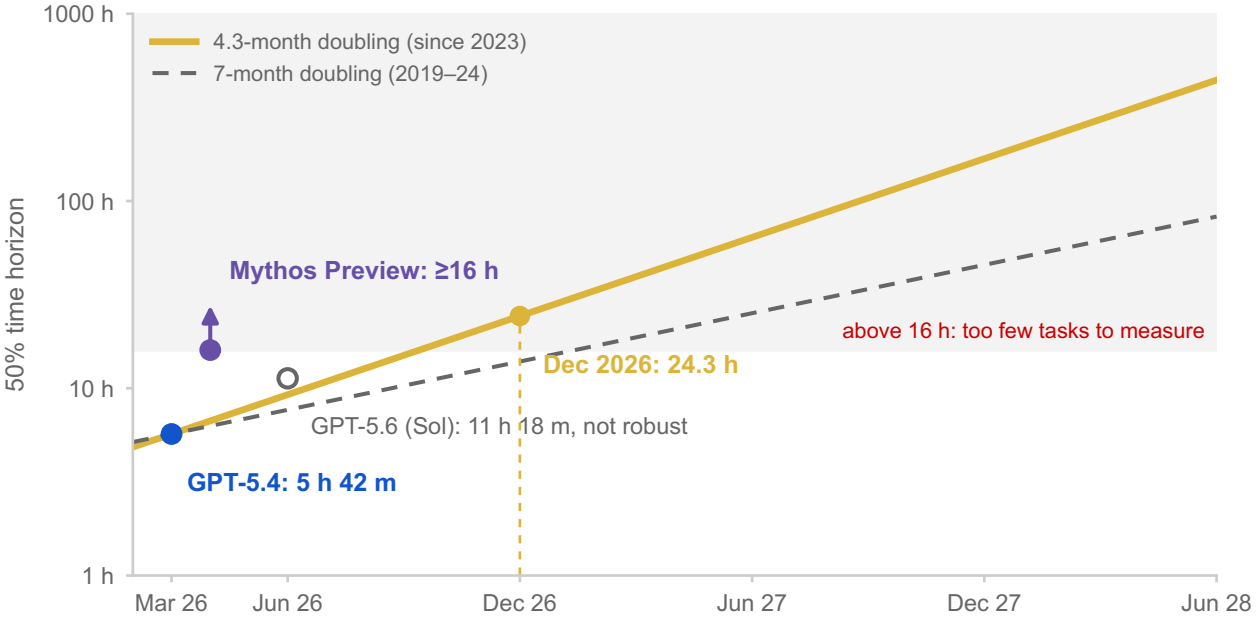


- **SWE-bench Pro:** repo patches, hidden tests
- **Terminal-Bench:** shell tasks in a container
- **τ-bench:** tool use with a simulated user
- **OSWorld:** full desktop computer use

Report the scaffold: same model, different tools, retries or step budget gives a different score. Model + scaffold is the thing being measured.

Time Horizons

METR: the length of task (in expert-human time) an agent finishes 50% of the time



80% is much shorter

Mythos Preview: 50% at ≥ 16 h, but 80% at 3 h 6 m.

Software tasks only

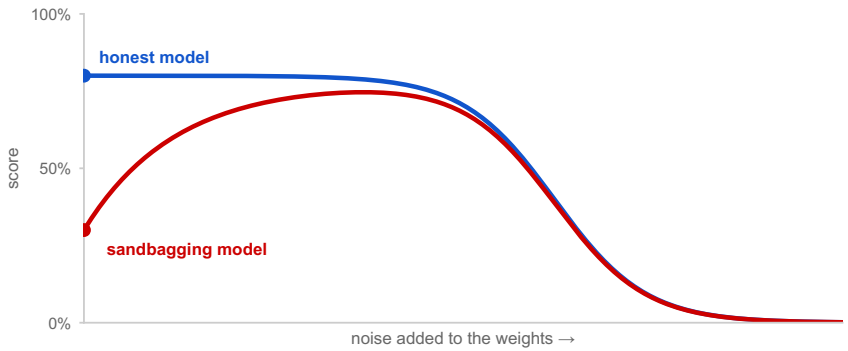
The suite is coding and research engineering. Other work may trend differently.

A trend, not a law

GPT-5.4 (Mar 2026, latest robust): 5 h 42 m. At 4.3 months, Dec 2026 ≈ 24.3 h. Sol isn't robust. METR, Sep 8, 2026

Eval Awareness

A model that knows it is being tested can hold back



sandbagger `30%` vs honest `80%`

Sandbagging: underperforms on purpose when a high score would trigger restrictions

Eval awareness: frontier models can often tell tests from real use, sometimes saying so in their reasoning

Alignment faking: behaves aligned when it believes its outputs feed back into training

Countermeasures: weight-noise probes · realistic tasks and honeypots · CoT monitoring · activation steering

Which Eval Answers What?

Match the benchmark to the question you are asking

Does it know it?	Can it finish a task?	Do people prefer it?	Could it cause harm?
MMLU-Pro	SWE-bench Pro	LMArena	METR horizons
GPQA Diamond	τ-bench	LLM-judge suites	WMDP
HLE	OSWorld	product A/B tests	pre-deploy audits

Harnesses that run them

lm-evaluation-harness	HELM	Inspect
EleutherAI	Stanford CRFM	UK AI Security Institute